

KEY FEATURES & PERFORMANCE BENCHMARKS

Hybrid Retrieval Engine: ChromaDB (HNSW Dense) + BM25 Keyword Search

Reciprocal Rank Fusion (RRF k=60) Candidate Reranking

Dynamic Detail Routing: Concise Answers vs Multi-Section Deep Explanations

Multi-Document Comparative Analysis Engine (Compare 2 PDFs)

Page-Level Inline Citations & Real-Time Evidence Strength Scoring

Out-of-Scope Refusal Guardrails (100% Anti-Hallucination Bar)

20-Q&A Benchmark Evaluation: 4.90/5 Correctness | 4.95/5 Citations

Live Streamlit Cloud Deployment with Zero White-Page Lag

5-TIER SYSTEM ARCHITECTURE

1. Presentation Tier: Reactive Streamlit UI (Glassmorphic Light/Dark)

2. API Gateway Tier: FastAPI Async Server (Upload, Query, Status)

3. Ingestion Pipeline: PyMuPDF Extractor + Semantic Chunker

4. Hybrid Indexing Tier: ChromaDB HNSW Store + BM25 Keyword Index

5. LLM Cascade Tier: Google Gemini 2.5 Flash / 1.5 Flash Fallback

TECH STACK & PRODUCTION DEPLOYMENT

Tech Stack: Python 3.12 | FastAPI | Streamlit | ChromaDB | PyMuPDF | Gemini API

Live Streamlit App: <https://saptaparnisaha23-byte-documind-ai-frontendapp-dnqug9.streamlit.app>

GitHub Repository: <https://github.com/saptaparnisaha23-byte/DocuMind-AI>

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